

"Crime Track Management System"

A DBMS MINOR PROJECT REPORT SUBMITTED TO

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in partial fulfillment for the award of degree of



Bachelor of Engineering in Information Science Engineering

Submitted By

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THE NATIONAL INSTITUTE OF ENGINEERING

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CERTIFICATE

Certifies that the project work titled "Crime Track Management System" is a work carried out by **PRANITH C GOWDA** of USN 4NI24IS136 of fourth semester BE in Information Science & Engineering in partial fulfilment for the requirements prescribed by The National Institute of Engineering, Autonomous Institution under Visvesvaraya Technological University, Belagavi. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated. The Project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the fourth semester in Data Base Management Systems Laboratory.

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ABSTRACT

Criminal records generally contain personal information about individuals along with identification details, FIR details and photographs. To identify criminals efficiently, proper data storage and retrieval systems are required. The Crime Track Management System is developed to simplify the process of maintaining criminal records using a computerized database system. Traditional manual record systems are time-consuming, difficult to maintain and may lead to data redundancy or data loss.

The system provides a structured digital solution for storing, updating, deleting and retrieving criminal and FIR records. It allows authorized users to manage criminal details, search records using Aadhaar number or FIR number and maintain FIR records efficiently. The project is implemented using JavaFX for the graphical user interface, JDBC for database connectivity and MySQL as the backend database.

The application includes modules such as Login, Dashboard, Add Criminal, Search Criminal, FIR Records, View Criminals, Operators and Reports. Overall, the system improves data accuracy, reduces paperwork, saves time and helps law enforcement authorities maintain criminal records in an organized and secure manner.

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CHAPTER 1:

INTRODUCTION

1.1 Background

The rapid increase in criminal activities and the complexity of managing law enforcement records manually has highlighted the urgent need for a computerized criminal record management system. Every investigative action, from filing an FIR to tracking a suspect, requires accurate and quickly retrievable information. Traditional paper-based systems suffer from data redundancy, slow retrieval and physical damage to records.

Despite the availability of general database tools, most police departments at the grassroots level still rely on registers and files that are difficult to search, update and secure. There exists a clear need for accessible, user-friendly software that can digitize criminal records and motivate operational efficiency. The CrimeTrack Management System addresses this gap by providing a centralized platform for managing, analyzing and retrieving criminal and FIR data.

1.2 Problem Statement

Police stations and law enforcement agencies have no straightforward mechanism to track, retrieve or update criminal records efficiently. Manual record systems are slow, prone to human error, and do not support quick searches when the number of records increases. There is no feedback mechanism that alerts officers to duplicate FIRs or missing criminal information.

1.3 Objectives

- Design and implement a relational database to store criminal profiles, FIR details, operator information and login credentials.
- Develop a JavaFX-based desktop application that accepts criminal record inputs and stores them using JDBC into a MySQL database.
- Provide search functionality using Aadhaar number or FIR number for quick record retrieval.
- Implement a dashboard and reports module to display statistical summaries.

- Store criminal photographs for easy visual identification
- To develop a computerized system for managing criminal records efficiently.
- To store and maintain criminal details such as name, alias, address, Aadhaar number, gender, and photographs in a centralized database.
- To provide a secure login system for authorized users and police operators.
- To simplify the process of adding, searching, updating, and viewing criminal information.
- To maintain FIR (First Information Report) records digitally for faster access and management.
- To reduce manual paperwork and improve data accuracy in police record handling.
- To provide quick criminal search functionality using Aadhaar number or criminal ID.

- To generate reports related to criminals, FIR cases, open cases, and closed cases.
- To improve data security and maintain consistency using DBMS concepts such as primary keys, foreign keys, and constraints.
- To create a user-friendly graphical interface using JavaFX for easy navigation and operation.
- To enable efficient handling of operator details and administrative activities.
- To minimize redundancy and maintain normalized database tables for better performance.
- To provide faster retrieval of information using SQL queries, joins, views, and indexes.
- To enhance overall crime record management efficiency in police departments.
- To demonstrate practical implementation of Database Management System concepts in a real-world application.

1.4 Scope

The application targets law enforcement personnel who manage criminal records, FIR filings and suspect tracking. The scope covers user authentication, criminal record management, FIR management, search operations, and report generation. The system is designed as a desktop JavaFX application backed by a MySQL database and is suitable for local deployment using Eclipse IDE and MySQL Workbench.

1.5 Organisation of the Report

Chapter 2 presents the proposed system architecture. Chapter 3 covers the overall system design including modules and workflow. Chapter 4 details the database design with ER diagrams and schema descriptions. Chapter 5 describes the implementation. Chapter 6 presents results and screenshots, followed by Conclusion, Future Enhancements and References.

CHAPTER 2:**PROPOSED SYSTEM****2.1 Existing System**

Existing criminal record management systems are predominantly manual, paper-based approaches that maintain physical registers for FIR records and criminal details. They lack dynamic searching, per-record image storage, centralized data access, or real-time status updates. None of them specifically address the need for quick digital retrieval using unique identifiers like Aadhaar number.

2.2 Proposed System

The proposed CrimeTrack Management System is a dynamic, database-driven desktop application that:

- Maintains a persistent record of criminal profiles per entry, linked to a unique criminal ID.
- Stores FIR details including crime type, date, and status, with a foreign key reference to the criminal table.
- Allows searching of criminal records using Aadhaar number or FIR number for rapid retrieval.
- Assigns case statuses (Open/Closed) to FIR records and displays summary statistics on the dashboard.
- Stores criminal photographs as image paths for visual identification.
- Manages operator accounts and provides role-based access to system modules.
- Generates statistical reports for criminal and FIR data.

2.3 Advantages of the Proposed System

- Real-time, searchable criminal record management using Aadhaar number.
- Normalized database schema ensures data consistency and eliminates redundancy.
- Dashboard statistics provide a quick operational overview for officers.
- Image path storage enables quick visual identification of criminals.
- Structured FIR management with open/closed status tracking improves case workflow.

2.4 Technology Stack

Component	Technology
Programming Language	Java
Frontend Framework	JavaFX
Database	MySQL
Database Connectivity	JDBC (Java Database Connectivity)

IDE	Eclipse IDE
Operating System	Windows 10/11

Frontend Technology – JavaFX

The frontend of the project is developed using JavaFX, which is a powerful Java library used for creating modern graphical user interfaces (GUI). JavaFX provides attractive UI components such as buttons, tables, forms, labels, images, and layouts which help in designing a professional-looking application. CSS styling is also used along with JavaFX to improve the visual appearance of the application screens such as Dashboard, Add Criminal, Search Criminal, FIR Records, Operators, Reports, and View Criminals.

JavaFX was chosen because it provides:

- Rich GUI components
- Better styling support using CSS
- Smooth event handling
- Easy integration with Java backend
- Platform-independent desktop application support

Backend Technology – Java

The backend logic of the project is implemented using Java. Java handles all the core functionalities of the application including:

- User authentication
- Database connectivity
- Record insertion and retrieval
- Search operations
- Report generation
- CRUD operations (Create, Read, Update, Delete)

Java was selected because it is:

- Object-oriented
- Secure and reliable
- Platform independent
- Easy to integrate with MySQL database
- Suitable for enterprise-level applications

CHAPTER 3:

SYSTEM DESIGN

3.1 System Architecture

The application follows a three-tier client-server-database architecture. The presentation tier consists of JavaFX FXML-based screens rendered using Scene Builder. The application tier is the Java backend running on the local machine, which handles routing, business logic, session management, and CRUD operations. The data tier is a MySQL relational database containing four core tables.

3.2 Module Description

3.2.1 Authentication Module

Handles user login with username and password validation against the LOGIN table. The login module verifies credentials and establishes an application session. Unauthorized access is blocked and redirected to the login screen.

3.2.2 Add Criminal Module

Users submit criminal details through the Add Criminal form. The backend extracts values for criminal number, name, alias, address, gender, Aadhaar number and image path, then inserts the data into the CRIMINAL table using a PreparedStatement to prevent SQL injection.

3.2.3 Search Criminal Module

The search module accepts an Aadhaar number or FIR number as input and executes a parameterized SELECT query against the CRIMINAL or FIR table. Results are displayed in a JavaFX TableView with all associated details.

3.2.4 Dashboard Module

The dashboard aggregates counts from the CRIMINAL, FIR and OPERATORS tables using SQL COUNT queries. Statistics such as total criminals, total FIRs, pending cases and operator count are displayed as summary cards on the dashboard screen.

3.2.5 FIR Records Module

The FIR Records module displays all FIR entries from the FIR table with fields including FIR number, criminal name, crime type, date and status. Officers can view, add and update FIR records through this module.

3.2.6 View Criminals Module

Displays all criminals stored in the CRIMINAL table along with their photographs. Each record shows the criminal's image path, name, alias, gender and Aadhaar number in a structured table layout.

3.2.7 Operators Module

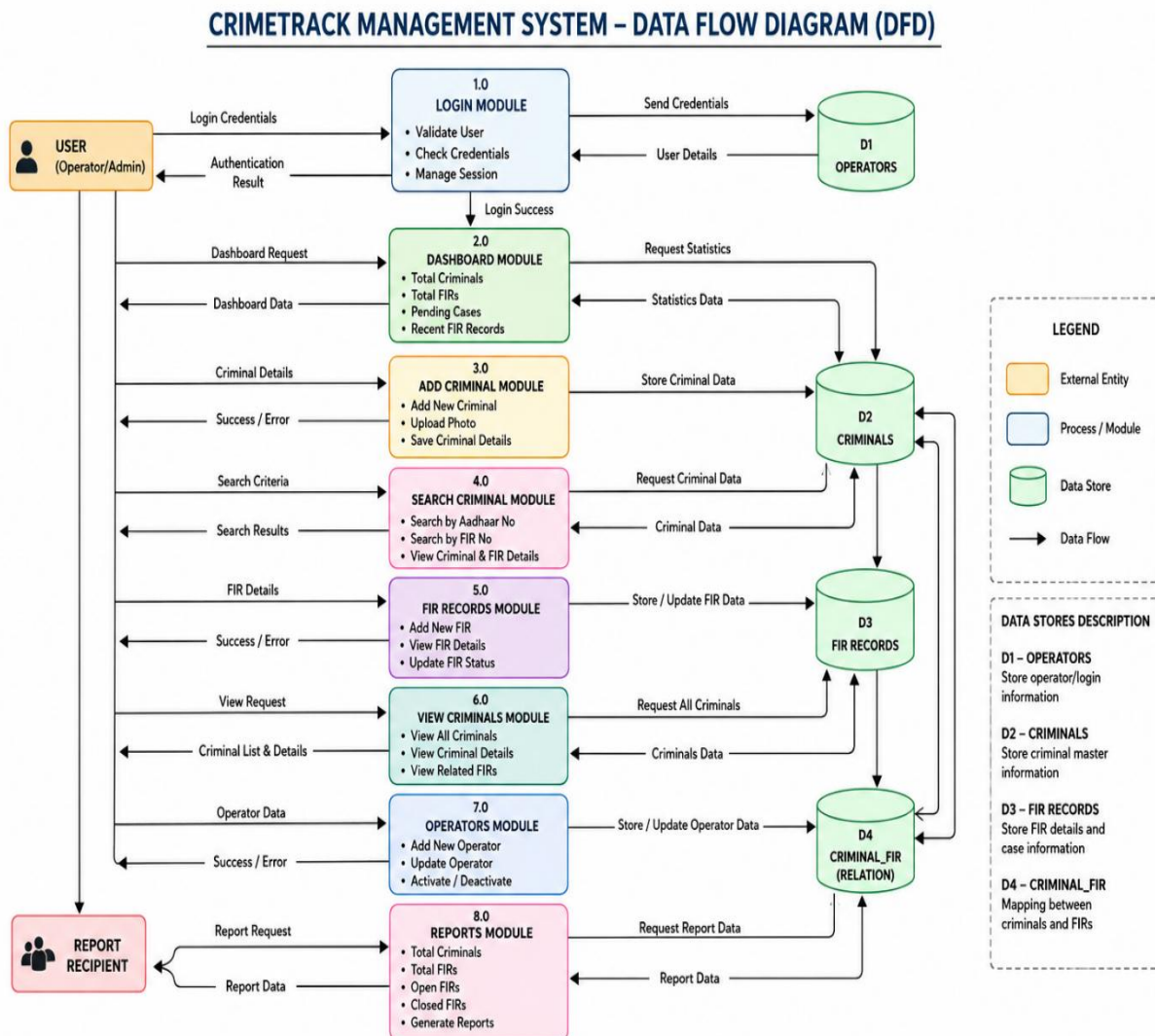
The Operators module manages operator accounts including operator ID, name, role, contact number and active/inactive status. This module allows administrators to add or deactivate operator accounts.

3.2.8 Reports Module

The Reports module queries the CRIMINAL and FIR tables to generate statistical summaries including total criminals registered, total FIRs filed, open cases and closed cases. These summaries are displayed in a report screen.

3.3 Data Flow

User logs in → Navigates to desired module via Dashboard → Submits form data → Java backend executes JDBC PreparedStatement → MySQL stores/retrieves data → Results displayed in JavaFX TableView or Dashboard cards.



CHAPTER 4:

DATABASE DESIGN

4.1 Entity-Relationship Overview

The database crimetrackdb (MySQL, utf8mb4) contains four tables: LOGIN, CRIMINAL, FIR and OPERATORS. Each table is defined with the InnoDB engine for full foreign-key constraint support and transactional integrity.

4.2 Table Descriptions

4.2.1 LOGIN Table

Column	Type	Description
username	VARCHAR(50) PK	Unique login username
password	VARCHAR(100) NOT NULL	User password

4.2.2 CRIMINAL Table

Column	Type	Description
criminal_id	INT AUTO_INCREMENT PK	Unique criminal identifier
criminal_no	VARCHAR(20)	Criminal record number
name	VARCHAR(100) NOT NULL	Criminal's full name
alias_name	VARCHAR(100)	Alias or nickname
address	VARCHAR(255)	Residential address
gender	VARCHAR(20)	Gender of criminal
aadhaar_no	VARCHAR(20) UNIQUE	Aadhaar identification number
image_path	VARCHAR(255)	Path to criminal's photograph

4.2.3 FIR Table

Column	Type	Description
fir_id	INT AUTO_INCREMENT PK	Unique FIR identifier
criminal_id	INT FK	Reference to CRIMINAL table
fir_no	VARCHAR(20)	FIR registration number

criminal_name	VARCHAR(100)	Name of accused criminal
crime	VARCHAR(150)	Type of crime committed
fir_date	DATE NOT NULL	Date FIR was filed
status	VARCHAR(20)	Open or Closed status

4.2.4 OPERATORS Table

Column	Type	Description
operator_id	VARCHAR(20) PK	Unique operator identifier
name	VARCHAR(100) NOT NULL	Operator's full name
role	VARCHAR(50)	Operator role designation
contact	VARCHAR(15)	Contact phone number
status	VARCHAR(20)	Active or Inactive status

4.3 Relational Schema

LOGIN(username PK, password)

CRIMINAL(criminal_id PK, criminal_no, name, alias_name, address, gender, aadhaar_no, image_path)

FIR(fir_id PK, criminal_id FK → CRIMINAL, fir_no, criminal_name, crime, fir_date, status)

OPERATORS(operator_id PK, name, role, contact, status)

4.4 Referential Integrity

Referential integrity is enforced between the FIR and CRIMINAL tables. The criminal_id column in the FIR table is a foreign key that references the criminal_id primary key in the CRIMINAL table. This constraint prevents FIR records from being created for non-existent criminal entries, ensuring data consistency across the database.

The foreign key constraint ensures that:

- An FIR record cannot exist without a valid criminal record.
- Only existing criminals can have FIR entries.
- Invalid or orphan FIR records are prevented.
- Data consistency and reliability are maintained across the database.

For example, if a criminal with `criminal_id = 5` exists in the CRIMINAL table, then only that valid ID can be inserted into the FIR table. If a non-existent `criminal_id` is entered while inserting FIR details, the database management system will reject the operation.

This integrity constraint helps maintain proper relationships between tables and avoids inconsistent data during insert, update, or delete operations.

4.5 Normalization

First Normal Form (1NF): All attributes contain atomic values with no repeating groups. Second Normal Form (2NF): All non-key attributes are fully functionally dependent on the primary key with no partial dependencies. Third Normal Form (3NF): No transitive dependencies exist, ensuring minimal redundancy. The database design satisfies normalization up to 3NF.

The database structure of the CrimeTrack Management System is properly normalized up to Third Normal Form (3NF). This ensures efficient storage of criminal and FIR records while maintaining consistency, integrity, and minimal redundancy in the database.

4.6 Indexes

Performance-critical indexes have been created to accelerate frequently executed queries:

- `idx_aadhaar` on `CRIMINAL(aadhaar_no)` — optimizes Aadhaar-based search queries.
- `idx_fir_no` on `FIR(fir_no)` — speeds up FIR number lookups.
- `idx_criminal_id` on `FIR(criminal_id)` — optimizes join queries between FIR and CRIMINAL tables.

4.7 Views

A View is a virtual table created using one or more SQL queries. Views are used to simplify complex queries, improve security, and provide customized representations of data without storing duplicate information physically in the database.

In the CrimeTrack Management System, views are used to display important criminal and FIR information efficiently. The following views are created in the project.

1. Criminal Summary View

This view displays important criminal details such as criminal number, name, alias name, address, gender, and Aadhaar number.

Query

```
CREATE VIEW criminal_summary_view AS
SELECT criminal_no, name, alias_name,
address, gender, aadhaar_no
FROM criminal;
```

Usage

```
SELECT * FROM criminal_summary_view;
```

2. Open FIR Cases View

This view displays all FIR records whose status is marked as “Open”. It helps police operators quickly identify active criminal cases.

Query

```
CREATE VIEW open_fir_view AS
SELECT fir_no, criminal_name,
crime, fir_date, status
FROM fir
WHERE status='Open';
```

Usage

```
SELECT * FROM open_fir_view;
```

3. Criminal and FIR Combined View

This view combines criminal details and FIR information using JOIN operation. It helps in generating reports and displaying complete criminal history.

Query

```
CREATE VIEW criminal_fir_view AS
SELECT c.criminal_no,
       c.name,
       c.aadhaar_no,
       f.fir_no,
       f.crime,
       f.fir_date,
       f.status
FROM criminal c
JOIN fir f
ON c.name = f.criminal_name;
```

Usage

```
SELECT * FROM criminal_fir_view;
```

Thus, views are useful in the CrimeTrack Management System for efficient data retrieval and reporting purposes.

CHAPTER 5:

IMPLEMENTATION

5.1 Development Environment

- Operating System: Windows 10/11
- Programming Language: Java (JDK 17+)
- Frontend: JavaFX with FXML
- Database: MySQL 8.0 (via XAMPP or MySQL Workbench)
- IDE: Eclipse IDE
- Connector: MySQL JDBC Connector (mysql-connector-java)

5.2 Project Structure

The project follows a standard Java project layout: src/ (main Java source files and controllers), resources/ (FXML layout files and CSS stylesheets), lib/ (JDBC connector JAR), and database/ (schema.sql containing all DDL statements).

5.3 Key Implementation Details

5.3.1 Database Connectivity

The database connection is established using JDBC's `DriverManager.getConnection()` method with the MySQL connection URL, username and password. A singleton `DBConnection` class is used to maintain a single active connection throughout the application lifecycle, preventing multiple redundant connections.

5.3.2 CRUD Operations

All database operations use `PreparedStatement` to safely bind parameters and prevent SQL injection. The `INSERT` statement is used for adding criminal and FIR records. `SELECT` with `WHERE` clause is used for search operations by Aadhaar number or FIR number. `UPDATE` modifies existing criminal details, and `DELETE` removes records by `criminal_id`.

5.3.3 Search Implementation

The search module accepts user input through a JavaFX `TextField` and executes a parameterized `SELECT` query: `SELECT * FROM criminal WHERE aadhaar_no = ?`. The result set is iterated and each record is loaded into an `ObservableList` that is bound to a JavaFX `TableView` for dynamic display.

5.3.4 Dashboard Data Aggregation

The dashboard controller executes four targeted SQL `COUNT` queries: one for total criminals, one for total FIRs, one for open cases (`WHERE status='Open'`), and one for total operators. Results are passed to JavaFX `Label` elements on the dashboard screen for display as summary statistics.

5.3.5 FIR-Criminal Join Query

To display FIR records alongside criminal details, an INNER JOIN is executed between the CRIMINAL and FIR tables on criminal_id. This join query retrieves the criminal name, crime type, FIR number, date and status in a single result set, which is then displayed in the FIR Records screen.

5.4 Frontend Implementation

The JavaFX frontend uses FXML layout files designed with Scene Builder for each module screen. CSS stylesheets provide a modern dark-themed sidebar navigation interface with styled buttons, table views and dashboard cards. Each screen controller extends a base JavaFX Controller class that manages JDBC connectivity and scene transitions

The application screens are implemented using JavaFX layouts and controls such as:

- Buttons
- Labels
- TextFields
- TableViews
- ImageViews
- ComboBoxes
- Dashboard cards
- Navigation panels

Each module of the system, such as Login, Dashboard, Add Criminal, Search Criminal, FIR Records, View Criminals, Operators, and Reports, is implemented as a separate JavaFX scene. These scenes are connected through controller classes that manage navigation and functionality.

The frontend design follows a modern dashboard-style layout with:

- Dark-themed sidebar navigation
- Styled action buttons
- Search bars
- Dashboard statistic cards
- Responsive tables
- Status indicators
- Criminal image previews

CSS stylesheets are used extensively to improve the appearance of the application. Custom CSS styling provides:

- Sidebar colors and hover effects
- Table formatting
- Button styling
- Dashboard card designs

- Font customization
- Border and layout styling

The application interface is organized using JavaFX layout containers such as:

- BorderPane
- VBox
- HBox
- GridPane
- StackPane

These layouts help maintain proper alignment and responsive screen arrangement.

Each JavaFX controller class is responsible for:

- Handling user actions
- Managing scene transitions
- Validating form inputs
- Displaying database records
- Performing CRUD operations through JDBC connectivity

The frontend communicates with the backend MySQL database using JDBC, enabling dynamic retrieval and display of criminal and FIR records.

Overall, the JavaFX frontend provides a professional and visually appealing interface for the CrimeTrack Management System while ensuring smooth interaction between the user and the database system.

CHAPTER 6:

RESULTS AND SCREENSHOTS

This chapter presents the key screens of the application, demonstrating the implemented functionality.

6.1 Login Screen



Description: The Login screen authenticates authorized users using a username and password. Invalid credentials display an error alert while valid credentials redirect the user to the Dashboard. Upon successful login, the user is redirected to the Dashboard module, which allows access to all functionalities of the CrimeTrack Management System.

6.2 Dashboard Screen

Description: The Dashboard displays summary statistics including total criminals registered, total FIRs filed, open cases and total operators, presented as styled information cards.

CRIME TRACK
Crime Management System

Dashboard

Add Criminal

Search Criminal

FIR Records

View Criminals

Operators

Reports

Logout

© 2024 CrimeTrack System

CrimeTrack - Dashboard

Dashboard

Welcome, Admin

Total Criminals

125

View Details →

Total FIRs

89

View Details →

Operators

12

View Details →

Pending Cases

23

View Details →

Recent FIR Records

View All

FIR No	Name	Crime	Date	Status
FIR00123	Ravi Kumar	Theft	17-04-2024	Open
FIR00122	Mohan Singh	Assault	16-04-2024	Open
FIR00121	Suresh Yadav	Fraud	16-04-2024	Closed
FIR00120	Ajay Verma	Robbery	15-04-2024	Open
FIR00119	Pawan Kumar	Cheating	14-04-2024	Open

6.3 Add Criminal Screen

CrimeTrack
Management System

Dashboard

Add Criminal

Search Criminal

FIR Records

View Criminals

Operators

Reports

Logout

CrimeTrack Management System

12:45:30 PM Admin

ADD CRIMINAL

Criminal ID

101

Criminal No

CRN2025001

Name

Ravi Kumar

Alias Name

Ravi

Address

No. 45, 2nd Cross, Bogadi, Mysuru - 570026, Karnataka

Gender

Male

Aadhaar No

9876 5432 1098

Image Path

C:\Images\criminal_101.jpg

Browse

Photo Preview

Save

Reset

Cancel

Total Criminals: 120

Logged in as: Admin

16 May 2025

Description: The Add Criminal screen allows entry of criminal details including name, alias, address, gender, Aadhaar number and photograph path. On submission, the record is inserted into the CRIMINAL table.

The uploaded image is displayed on the screen using JavaFX ImageView component, which helps operators visually identify criminals easily. This module simplifies criminal data entry and improves record management efficiency.

Image Insertion Explanation

The criminal image is inserted using JavaFX FileChooser. When the user clicks the Upload Image button, the system opens the file explorer and allows image selection. The selected image path is stored in the `image_path` column of the CRIMINAL table.

Example image path stored:

`images/criminal1.jpg`

The image is then displayed dynamically in the application using JavaFX ImageView component.

6.4 Search Criminal Screen

SEARCH CRIMINAL

Search By: Aadhaar No (selected) | Enter Aadhaar No: 987654321098 | Search

Search Results

 Criminal ID : 101 | Criminal No : CRN2025001 | Image Path : C:\Images\criminal_101.jpg
Name : Ravi Kumar | Total FIRs : 2
Alias Name : Ravi | Created Date : 10-05-2025
Address : No. 45, 2nd Cross, Bogadi, Mysuru - 570026, Karnataka | Last Updated : 15-05-2025
Gender : Male | Aadhaar No : 9876 5432 1098 | [View FIR Records](#)

FIR Records Summary

FIR ID	FIR No	Crime	FIR Date	Status
201	FIR2025001	Theft	12-04-2025	Active
202	FIR2025003	Assault	02-05-2025	Under Investigation

Total Criminals: 120 | Logged in as: Admin | 16 May 2025

Description: The Search screen accepts an Aadhaar number or FIR number and retrieves matching criminal records from the database, displaying them in a structured table view.

6.5 FIR Records Screen

FIR No	Name	Crime	Date	Status
FIR00123	Ravi Kumar	Theft	17-04-2024	Open
FIR00122	Mohan Singh	Assault	16-04-2024	Open
FIR00121	Suresh Yadav	Fraud	16-04-2024	Closed
FIR00120	Ajay Verma	Robbery	15-04-2024	Open
FIR00119	Pawan Kumar	Cheating	14-04-2024	Open
FIR00118	Deepak Rao	Burglary	14-04-2024	Closed
FIR00117	Anil Gupta	Kidnapping	13-04-2024	Open
FIR00116	Naveen Shetty	Assault	12-04-2024	Open
FIR00115	Rahul Das	Cheating	11-04-2024	Closed
FIR00114	Vikram Jain	Forgery	10-04-2024	Open

Total Records: 89

Description: Displays all FIR records with their associated criminal name, crime type, FIR date and open/closed status. Officers can add new FIR records from this screen.

6.6 View Criminals Screen

Criminal No	Photo	Name	Alias	Aadhaar No	Action
CR00125		Ravi Kumar	Ravi	123456789012	
CR00124		Mohan Singh	Mohan	987654321098	
CR00123		Suresh Yadav	Suresh	456789123456	
CR00122		Ajay Verma	Ajay	789123456789	
CR00121		Pawan Kumar	Pawan	321654987123	

Total Records: 125

Description: Displays all criminal records in a tabular format with image path, name, alias, Aadhaar number and action buttons for update or delete operations.

6.7 Operators Screen

Operator ID	Name	Role	Contact	Status	Action
OP001	Admin Officer	Admin	9876543210	Active	
OP002	Police Operator	Operator	9876501234	Active	
OP003	Record Manager	Operator	9123456789	Active	
OP004	FIR Handler	Operator	9988776655	Inactive	
OP005	Data Entry Operator	Operator	9001234567	Active	

Total Operators: 5

Description: Displays operator accounts with their ID, name, role, contact number and active/inactive status for administrator management.

6.8 Reports Screen

Metric	Value
Total Criminal Records	125
Total FIR Records	89
Open FIR Cases	58
Closed FIR Cases	31

Report Generated On: 16-05-2025 12:45 PM

[Generate Report](#)

The Reports screen displays summarized information about criminal records and FIR case statistics using dashboard cards and report tables. It helps administrators monitor total criminals, open FIRs, closed FIRs, and overall crime-related data efficiently

6.9 Database Tables

6.8 Database Tables

Description: Shows the MySQL database tables (CRIMINAL, FIR, LOGIN, OPERATORS) and their records stored in the **crimetrackdb** database.

SCHEMAS

Filter objects

crimetrackdb

Tables

criminal

fir

login

operators

Views

Stored Procedures

Functions

sys

world

criminal - Table

Filter Rows

Edit

Export/Import

criminal_id	criminal_no	name	alias_name	address	gender	aadhaar_no	image_path
1	CR001	Ravi Kumar	Ravi	Delhi	Male	123456789012	image1.jpg
2	CR002	Amit Singh	Amit	Mumbai	Male	234567890123	image2.jpg
3	CR003	Sanjay Das	Sanjay	Kolkata	Male	345678901234	image3.jpg
4	CR004	Neha Sharma	Neha	Jaipur	Female	456789012345	image4.jpg
5	CR005	Vikram Patil	Vicky	Pune	Male	567890123456	image5.jpg
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

criminal 1

login - Table

Filter Rows

Edit

Export/Import

login_id	username	password	role
1	admin	admin123	Administrator
2	operator1	op123	Operator
3	operator2	op456	Operator
NULL	NULL	NULL	NULL

login 1

fir - Table

Filter Rows

Edit

Export/Import

fir_no	criminal_id	criminal_name	crime	fir_date	status	
1	FIR001	1	Ravi Kumar	Theft	2024-01-15	Open
2	FIR002	2	Amit Singh	Robbery	2024-02-20	Open
3	FIR003	3	Sanjay Das	Assault	2024-03-10	Closed
4	FIR004	1	Ravi Kumar	Fraud	2024-03-25	Open
5	FIR005	4	Neha Sharma	Kidnapping	2024-04-05	Open
6	FIR006	5	Vikram Patil	Cyber Crime	2024-04-18	Closed
NULL	NULL	NULL	NULL	NULL	NULL	NULL

fir 1

operators - Table

Filter Rows

Edit

Export/Import

operator_id	name	phone	email	address	status
1	Rajesh Verma	9876543210	rajesh@crimetrack.com	Delhi	Active
2	Sunita Sharma	9123456780	sunita@crimetrack.com	Mumbai	Active
3	Amit Yadav	9988776655	amit@crimetrack.com	Kolkata	Inactive
NULL	NULL	NULL	NULL	NULL	NULL

operators 1

crimetrackdb

4 tables

Read Only

Description: Shows the MySQL database tables (CRIMINAL, FIR, LOGIN, OPERATORS) and their records stored in the crimetrackdb database.

The **CRIMINAL** table stores complete criminal information including criminal number, name, alias name, address, gender, Aadhaar number, and image path. This table acts as the primary source for criminal record management.

The **FIR** table stores First Information Report details such as FIR number, crime type, FIR date, status, and associated criminal details. It maintains the relationship between criminal records and FIR cases using primary and foreign key constraints.

The **LOGIN** table is used for user authentication and stores usernames, passwords, and user roles for secure access to the system. Only authorized users can access the CrimeTrack application through the login module.

The **OPERATORS** table stores details of police operators and administrators including operator ID, name, role, contact information, and account status. This helps manage system users efficiently.

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CONCLUSION

The CrimeTrack Management System successfully demonstrates how database management systems, desktop application development and law enforcement operations can be integrated to build a practical, user-centric criminal record management tool. The project implements a fully normalized four-table MySQL schema that cleanly separates login credentials, criminal profiles, FIR records and operator accounts, achieving data integrity through foreign-key constraints and performance through strategic indexing.

The JavaFX frontend efficiently handles user authentication, criminal record management, FIR filing, dashboard aggregation and report generation. The JDBC-powered backend provides reliable and secure CRUD operations that make complex database interactions accessible through a user-friendly interface. The search module using Aadhaar number has proven effective in providing rapid criminal record retrieval for investigative workflows.

This project reaffirms that digitization is the first step toward efficient law enforcement record management, and technology is the most scalable vehicle for delivering that efficiency at the department level.

FUTURE ENHANCEMENTS

- **Face Recognition Integration:** Implement OpenCV-based face recognition to identify criminals from uploaded photographs automatically.
- **Fingerprint Verification:** Integrate biometric fingerprint scanning for secure officer authentication and criminal identification.
- **Cloud Database Support:** Migrate the MySQL database to a cloud platform such as AWS RDS or Google Cloud SQL for remote multi-station access.
- **PDF Report Generation:** Implement automated PDF generation for criminal profiles and FIR summaries using iText or Apache PDFBox.
- **Role-Based Access Control:** Expand the authentication system to support multiple roles such as Administrator, Investigator and Read-Only Viewer with granular permission sets.
- **Mobile Application:** Develop an Android companion app that allows officers to search criminal records from field devices.
- **Online FIR Registration:** Build a citizen-facing web portal that allows online FIR registration and complaint tracking.
- **AI-Based Criminal Identification:** Integrate machine learning models to predict crime patterns and assist in suspect profiling based on historical FIR data.

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